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Large-scale micro-aerobic digestion studies at municipal water resource recovery facilities for process-integrated biogas desulfurization

Norbertus Joannes Richardus Kraakman [a,b,*], Israel Diaz [b], Maria Fdz-Polanco [b], Raul Muñoz [b]

[a] Jacobs Solutions, One Glass Wharf, Bristol BS2 0EL, United Kingdom
[b] Institute of Sustainable Processes, University of Valladolid, Doctor Mergelina s/n, 47011 Valladolid, Spain

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ABSTRACT

Biogas generated from anaerobic digestion processes typically requires gas conditioning including hydrogen sulfide removal before it is used on-site to generate electricity and heat or before upgrading to biomethane for gas pipeline distribution or vehicle fuel. Experiments of process-integrated micro-aeration in three different large-scale sludge digesters (effective volume 2450 m^3, 3250 m^3 and 3300 m^3) were undertaken at municipal wastewater treatment plants and compared with large-scale applications reported in literature. This work discusses critical process parameters, advantages, and potential limitations of micro-aerobic digestion in comparison with other commonly used biogas desulfurization methods. It shows that high H_2S reduction in the biogas can be obtained and that the H_2S reduction is dependent not only on the air injection rate and location, but also on the digestion activity identified by the volumetric biogas production. Reliability testing of typical micro-aerobic upset scenarios and long-term micro-aerobic digester operation as part of this study provided deeper insights in the MA process that could also provide valuable input to what operational, maintenance and safety measures should be considered when applying micro-aeration in large-scale digesters. The large-scale micro-aerobic digestion trials demonstrated that the hydrogen sulfide concentration in the biogas can be controlled using minimal plant upgrades, contributing to the sustainability and economic efficiency of the energy recovery process of waste sludge digestion at water resource recovery facilities.

1. Introduction

Anaerobic digestion (AD) is extensively used to process organic wastes such as sewage sludge and source segregated organics (SSO), including agricultural waste materials and the organic fraction of municipal solid waste. Anaerobic digestion is more commonly used in recent years because of its potential to generate renewable energy from diverse feedstocks [1]. AD may also be used to divert organic wastes from landfills, thus minimizing the operational costs and environmental impacts of landfills. Moreover, AD is increasingly used as the core waste-to-renewable-energy process of sludge management at large municipal water resource recovery facilities (WRRFs), while it stabilizes the waste sludge, reduces potential pathogens, and turns the sewage sludge into a natural fertilizer [2].

The produced biogas typically requires conditioning before it is used to generate energy, typically in internal combustion engines, turbines and boilers, or before its upgrading to biomethane for gas pipeline distribution or use as a vehicle fuel. Elevated concentrations of hydrogen sulfide (H_2S) in the biogas are often a limitation for downstream processing of the biogas as it creates maintenance related corrosion problems in combustion engines, occupational health issues and the release of SO_x in flue gases, which entails harmful environmental effects and are subject to air permit restrictions [3]. The H_2S concentration in biogas varies from a few hundred to up to 10,000 ppm_v, depending on the sulfate inlet concentration and the amount of bioavailable sulfur compounds in the feedstock. The maximum H_2S concentration when biogas is used as fuel in cogeneration engines for combined heat and power (CHP) production is typically <100 ppm_v, depending on the manufacturer, and <5 ppm_v for biomethane gas pipeline distribution or use as a vehicle fuel [4,5].

Elevated H_2S concentrations in biogas are usually obtained by iron salts dosing into the digester or by applying conventional physical/chemical end-of-pipe biogas desulfurization technologies (i.e., wet scrubbing using water/caustic/hypochlorite or iron oxide–based adsorbents). These methods are relatively expensive due to ongoing operational costs and may cause high environmental impacts [6]. Cost-

* Corresponding author at: Jacobs Solutions, One Glass Wharf, Bristol BS2 0EL, United Kingdom.
E-mail address: bart.kraakman@jacobs.com (N.J.R. Kraakman).

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effective and more environmentally friendly alternatives have recently emerged that include biotechnology (i.e., aerobic or anoxic biotrickling filtration) or physical-chemical combinations with biotechnology (i.e., caustic scrubbing followed by biological oxidation) as discussed by Dumont [7]. However, these technologies are still hampered by their high investment costs and their generated byproducts that need further processing or disposal (i.e., sulfuric acid and/or elemental sulfur slurry).

A simple and economical alternative is a process-integrated biochemical method of removing H_2S from the biogas called micro-aeration (MA) and involves the dosing of a small amount of air or oxygen during the digestion process. The MA process takes place inside the digester and the bacteria responsible for sulfide oxidation are already present in the feed sludge. Micro-aerobic conditions during anaerobic digestion have been studied at bench-scale and proven to be an effective method for reducing H_2S concentrations in the produced biogas [8–10]. MA has been researched and used in agricultural biogas plants, but rarely tested at the typical larger sludge digesters at municipal WRRFs. It has been demonstrated in Europe in a few large-scale sludge digesters at WRRFs, with H_2S removal efficiencies of >90 % in most cases [11,12].

MA may also improve the degradation of organic matter (volatile suspended solids) and enhance methane production [13]. In addition, there is evidence that MA can enhance the stability of the digestion process, lower the foaming potential of the digested sludge, improve sludge dewaterability and reduce the chemical oxygen demand (COD) and dissolved sulfide concentration in the digested sludge centrate [13–16]. The potential drawbacks of MA include methane dilution (when relatively large amounts of air/oxygen are injected), and an accumulation of elemental sulfur in downstream gas pipes or flame arrestors [17]. Inhibition of the methanogenesis may also occur when too much air/oxygen is injected [18,8,19].

A comprehensive process understanding is necessary under larger-scale conditions, which can facilitate the scale-up and optimization of the MA process to obtain high quality biogas, maintain a stable digestion process, and eliminate unexpected maintenance and operational costs of the micro-aerobic digestion process. Operational and design parameters such as the amount of air dosed, the air dosing location and the liquid conditions, have shown to be important in bench-scale studies [18–24]. Nevertheless, additional digester design and operating parameters may be critical when scaling-up the MA process to successful implementation in large-scale digestion processes. This can include the type of organic feed material, the reactor type (e.g., internal configuration and volume, headspace volume, and/or reactor sludge mixing method) and the operational conditions.

The aim of this study is to investigate the biogas desulfurization performance in three different large-scale sludge digesters using MA. In addition, the impact of sudden changes in micro-aeration rate and typical digestion upset conditions are studied to better understand the MA process reliability. The findings are compared with bench-scale and other large-scale studies reported in literature to discuss critical process parameters and potential limitations in comparison with other commonly used biogas desulfurization methods, as well as opportunities to be considered when scaling-up MA for the waste-to-energy digestion process of sludge management at large municipal WRRFs.

2. Materials and methods

MA technology was studied over a period of multiple months in large-scale digesters at WRRFs in North America all different in terms of reactor size, organic loading rate, H_2S concentration in the biogas and/or sludge mixing method. The waste sludge fed into the digesters was in all cases a mixture of primary sludge (PS) and waste activated sludge (WAS). The H_2S concentrations in the biogas in the period immediately before or after the MA test were approximately 5500 ppm_v, 1000 ppm_v and 2250 ppm_v for Case Studies #1, #2 and #3, respectively. Sludge mixing in the digesters was done either by means of sludge recirculation through pumping (Case Study #1 and #3) or through biogas recirculation (Case Study #2). The average organic loading rate (OLR) was 2.1, 3.3 and 2.4 $kg_{VS}\ m_r^{-3}\ d^{-1}$ during the evaluation period for Case Studies #1, #2 and #3, respectively.

2.1. Case Study #1

MA technology was implemented over a period of ten weeks in a large-scale 20-meter diameter digester with an effective volume of 2450 m^3. The waste sludge fed into the digester was about 105 $m^3\ d^{-1}$, after gravity thickening. The feed sludge was a mixture of PS and WAS, in a volumetric ratio of approximately 1 to 1, with a TS concentration of about 6.0 % (*w*/w) and a VS content that was ~79 % of TS. The sludge was digested under mesophilic conditions at a hydraulic retention time of approximately 22 days.

Mechanical sludge recirculation was used for sludge mixing inside the digester with an average recirculation flowrate of 850 $m^3\ h^{-1}$, which resulted in an average sludge turn-over rate of ~3 h. The recirculated sludge was returned to the digester either near the bottom of the digester reactor or near the top of the digester just below the sludge surface using alternating valves (the discharge location was switched every 4 h from top to bottom). The OLR was ~2.1 $kg_{VS}\ m_r^{-3}\ d^{-1}$ during the evaluation period. The H_2S concentration in the biogas prior to the large-scale MA period ranged between 5000 and 6000 ppm_v.

During the MA tests ambient air (compressed service-air) was injected with fine bubble air injection nozzles in the sludge recirculation pipe downstream of the sludge recirculation pump (see Supplementary Fig. S-1 for the developed and implemented test set-up). The biogas H_2S concentration was monitored under different air dosing conditions. In addition, the impact of micro-aeration on the volatile solids reduction (VSR) was measured. The impact of sudden changes in micro-aeration rate on the biogas H_2S concentration was also evaluated. The flowrate of air injected and the H_2S concentration in the biogas were measured daily. Digested sludge samples were taken daily for analyses and sludge feed, sludge recirculation rate and biogas production were measured continuously.

2.2. Case Study #2

MA technology was also studied over a period of six months in a large-scale digester at WRRF in a 21-meter diameter digester (effective volume of 3250 m^3). The waste sludge flowrate fed into the digester was ~180 $m^3\ d^{-1}$. The digester was fed with PS and WAS (volumetric ratio of approximately 1 to 1), which were co-thickened in dissolved air flotation tanks (DAFTs). The sludge exhibited a TS concentration of ~6.0 % (*w*/w) and a VS content that was ~87 % of TS. The sludge was digested under mesophilic conditions with a hydraulic retention time of about 18 days, with the digester being mixed via gas recirculation. The H_2S concentration in the biogas prior to the large-scale MA period ranged between 1000 and 1200 ppm_v.

The air injection set-up during the large-scale micro-aerobic digestion tests in shown in Fig. S-2. Ambient air (compressed service-air) was initially injected in the recirculating sludge heating line just downstream of the sludge feed point of the digester. The large-scale pilot test continued injecting the air in the bottom of the digester using an existing 25-milimeter pipe after isolating one of the twelve biogas feeder pipes from the digester gas mixing system. Finally, the desulfurization performance was monitored under different micro-aerobic air dosing conditions during continuous and discontinuous (1 h on–1 h off) gas recirculation.

The amount of air injected and the H_2S concentration in the biogas were measured daily. Digested sludge samples were taken daily for analyses and sludge feed, sludge recirculation rate and biogas production were measured continuously. Sulfur analyses were performed of the feed sludge and the digested sludge of the micro-aerated digester during Case Study #2 to generate a sulfur mass balance of the micro-aerobic digestion process. The sulfur analyses were performed during the

period when micro-aeration air was injected in the sludge recirculation line during continuous gas mixing. In addition, the oxidation reduction potential (ORP) of the sludge in the digester was measured over a period of approximately one month.

2.3. Case Study #3

Finally, MA technology was studied over a period of about three months in a third large-scale digester (16-meter diameter with an effective volume of 3300 m^{-3}). The waste sludge flowrate fed into the digester was ~111 $m^3\ d^{-1}$, after gravity thickening. The feed sludge was a mixture of PS and WAS, in volumetric ratio of approximately 1 to 1, with a TS content of ~4.8 % (*w*/w) and a VS content that was ~87 % of TS. The sludge was digested under mesophilic conditions designed under a hydraulic retention time of approximately 30 days.

Mechanical sludge recirculation was used for sludge mixing inside the digester with an average recirculation flowrate of 749 $m^3\ h^{-1}$, which resulted in the sludge in the digester being turned over through recirculation pumping on average once every about 4.4 h. The recirculated sludge was taken from the bottom middle of the digester and returned near the top of the digester at ~2 m below the sludge surface at four evenly distributed locations 1 m from the wall to create a continuous swirl of digesting sludge inside the digester. The OLR was ~2.4 $kg_{VS}\ m_r^{-3}\ d^{-1}$ during the evaluation period. The average H_2S concentration in the biogas prior to the large-scale MA test ranged between 2000 and 2500 ppm_v.

Ambient air (compressed service-air) was injected with fine bubble air injection nozzles in the sludge recirculation pipe downstream of the sludge recirculation pump. Fig. S-3 shows the air injection set-up that was implemented during the large-scale digester testing. The desulfurization performance was monitored under different micro-aerobic air dosing conditions.

The amount of air injected and the H_2S concentration in the biogas were measured daily. Digested sludge samples were taken daily for analyses and sludge feed, sludge recirculation rate and biogas production were measured continuously. The flowrate of air injected and the H_2S concentration in the biogas were measured daily. Digested sludge samples were taken daily for analyses while sludge feed, sludge recirculation rate and biogas production were measured continuously.

2.4. Process upsets responses

Different possible process changes and potential process upset scenarios were investigated to gain an improved understanding of the reliability of the process-integrated MA under large-scale operations. Reliability is the combination of robustness and resilience, where robustness reflects the capacity of a system to maintain its functionality when subjected to changes. The resilience is the rate at which a system returns to its original state after being perturbed. Several digestion or micro-aeration process changes or upset scenarios were investigated, including a sudden MA-air injection stop (all Case Studies), MA-air injection rate overdose (Case Study #1), intermittent sludge gas mixing, a gas mixing blower failure (Case Study #2) and a sludge recycle pump failure directly followed by a digester feed pump stop (Case Study #3).

2.5. Analytical procedures

Sludge characteristics such as pH, alkalinity, total solids (TS) content and volatile solid (VS) content were measured daily using standard methods [25], while biogas production and sludge feeding rate were measured continuously using electromagnetic flowmeters. The air injection rate was measured using a flowmeter (Dwyer, RMD 0.5 – 600 scfh, 100 psi) and daily recorded. The H_2S concentration in the biogas was measured daily using either Dräger tubes or a handheld biogas analyzer (BIOGAS 5000, QED Environmental Systems, Inc., USA) and verified by taken biogas samples with silonite canisters for Fixed Gas lab analysis by standard method EPA 3C and the total reduced sulfur analyses by standard method ASTM-D-5504.

For sulfur analyses, the sludge samples were collected by filling a plastic bottle quickly and completely without getting sludge in much contact with air. Concentrated sodium hydroxide was added to the samples to raise the pH to a minimum of 8.5 and directly centrifuged to obtain a liquid fraction and a solid fraction. The liquid and solids fraction samples were stored in the fridge and shipped cold the next day to an analytical lab (ALS Environmental, Tucson, Arizona). Both the liquid and solid fractions were analyzed on different sulfur constituents (sulfides, sulfate and total sulfur). The sulfide concentration was analyzed according to EPA standard method M9030, the sulfate concentration was analyzed in accordance with EPA standard M9056 and the total sulfur concentration was analyzed as per ASTM methods D1552/D4239.

ORP measurements were carried out using a handheld ORP meter (Hach HQ40d multimeter) during Case Study #2. In addition, an online ORP meter (Hach pHD sc with platinum electrode in stainless steel flow cell) was also tested in the bypass line of the sludge recirculation pump using a special ORP probe injection piece from the manufacturer.

3. Results

3.1. Case Study #1

After introducing the air into the sludge on day 0, the H_2S concentration in the biogas slowly started to drop over about a week (Fig. 1). The initial air dosing rate was 4 $m^3\ h^{-1}$, which was increased to 8 $m^3\ h^{-1}$ after 3 weeks. Air dosing was further increased in several smaller steps over a period of 5 weeks to about 14 $m^3\ h^{-1}$. At day 0, two fine bubble air injection nozzles were used, at day 26 three fine bubble air injection nozzles were used and finally at day 56 all the six installed air nozzles were in operation.

The H_2S concentration in the produced biogas ultimately dropped over a period of 8 weeks by >80 % from about 5000 ppm_v to <1000 ppm_v. After stopping the MA on day 60, the H_2S concentration increased back to about 5000 ppm_v. The presence of MA in the anaerobic digestor showed no negative effects on organic matter removal, or biogas production when the amount of air injected was limited to less than about 13.6 $m^3\ h^{-1}$. Biogas was produced under mesophilic conditions and its production rate was on average 4508 $Nm^3\ d^{-1}$ (average of three digesters using one biogas flowmeter installed in the common outlet gas header) or 0.8 m^3 per kg_{VS} under a digester loading rate of 2.14 $kg_{VS}\ m_r^{-3}\ d^{-1}$, which didn't change during the test period of 8 weeks.

The pH of the sludge was 7.3 ± 0.15 without any pH control, while the operational temperature was kept on average at 33 ± 4 °C. The digested sludge during the test period had an average TS content of 2.5 ± 0.2 % (*w*/w). While methane production rate remained basically unchanged when introducing MA, the VS/TS ratio of the digested sludge dropped in this case by a few percent from about 66.6 % without MA to about 63.6 %.

3.2. Case Study #2

Directly after initiating the air supply into the digester's sludge recirculation line, the H_2S concentration in the biogas dropped within one week from about 1000 ppm_v to about 500 ppm_v (Fig. 2). After three weeks at an aeration rate of 2.5 $m^3\ h^{-1}$, which entailed an H_2S concentration of ~400 ppm_v, the air injection rate was slowly increased stepwise. This ultimately resulted in a biogas H_2S concentration of ~200 ppm_v when 5.7 $m^3\ h^{-1}$ of air was injected.

The testing was continued by dosing air injected in the bottom of the digester compared to the previous phase where the air was injected into the sludge recirculation line. During this test, the H_2S concentration in the biogas dropped within three weeks from about 1000 ppm_v to 400 to 500 ppm_v. After 3 weeks (day 21), the air injected was increased, which did not result in a further drop of the biogas H_2S concentration. At day 53, the dosing of air was further increased, which resulted in a H_2S

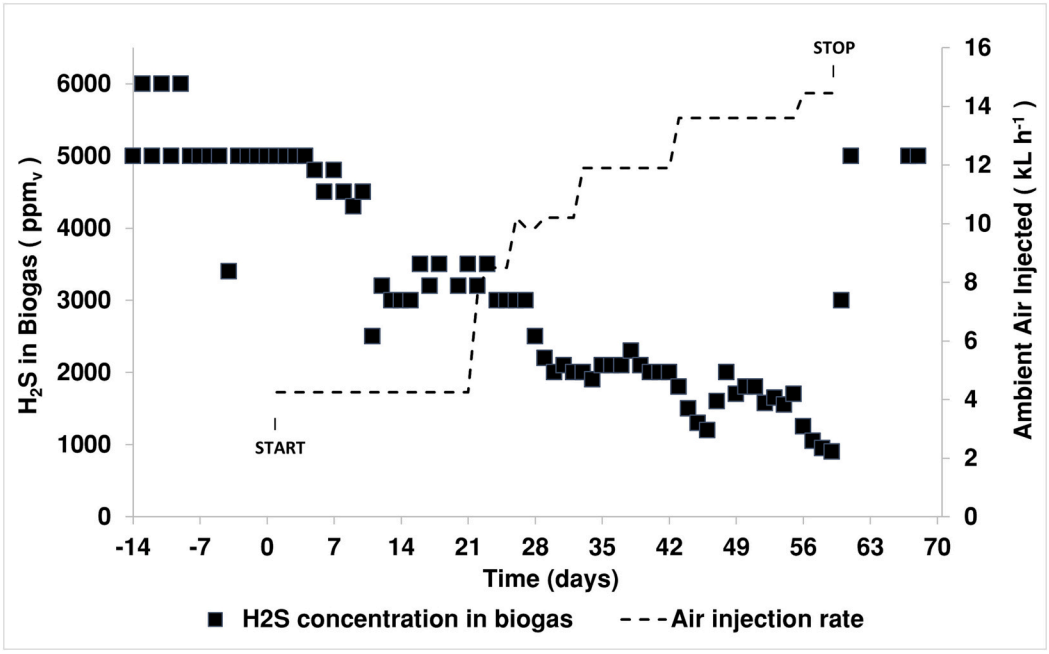


Fig. 1. Hydrogen sulfide concentration and ambient air injected in the biogas prior, during and after micro-aeration in Case Study #1.

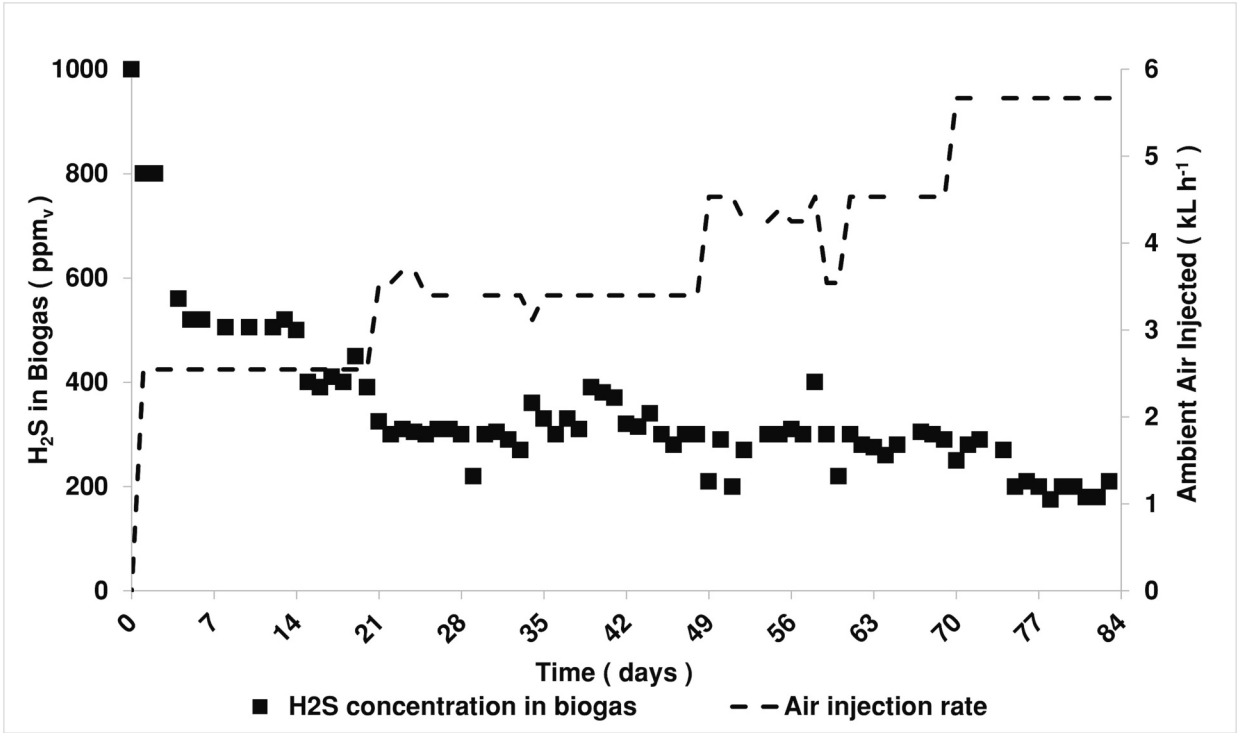


Fig. 2. Hydrogen sulfide concentration in biogas after starting micro-aeration in Case Study #2 with air injected in the sludge recirculation pipe.

concentration only slightly dropping further to around 400 ppm_v when 4.5 $m^3\ h^{-1}$ of air was injected. Ten days later (Day 63), the amount of air was further increased to 5.0 $m^3\ h^{-1}$, which did not result in a further reduction in the biogas H_2S concentration. The lowest H_2S concentration in the biogas measured was 330 ppm_v (data not shown).

An additional large-scale test was undertaken to investigate the performance of air injection at the two previous locations combined with intermittent sludge mixing. The digester was supplied with air prior to the start of this test phase at the bottom of the digester and additional air was injected into the sludge recirculation line using the spool piece with the three air injection nozzles as previous described. The gas mixing regime was changed from continuous mixing to intermittent mixing (1 h on/1 h off) shortly after day 8. During this test phase, the air flow rate injected in the sludge recirculation pipe was maintained constant at 1 $m^3\ h^{-1}$, while the amount of air injected in the bottom of the digester was about 5 $m^3\ h^{-1}$ and slightly increased over time. The average biogas H_2S concentration did not reduce further (compared to the previous test phase where MA air was only injected in the bottom of

the digester) and during the following weeks was in general higher than the 400 ppm_v prior to the start of this test phase. The H_2S concentration was less consistent and in general fluctuated between 350 and 750 ppm_v (data not shown). H_2S measurements were taken when the sludge mixing was on, for consistency. Further increasing the amount of air injected did not result in a decrease of biogas H_2S concentration.

During the test period, the biogas was produced under mesophilic conditions at a rate of about 1639 $Nm^3\ d^{-1}$ with a OLR of ~3.3 $kg_{VS}\ m_r^{-3}\ d^{-1}$. The pH of the sludge was 7.27 ± 0.07, while the operational temperature was kept on 35.8 ± 0.6 °C. The ORP measured during Case Study #2 in the digested sludge samples ranged between −515 and −350 mV, which agreed with the expected values of the anaerobic broth in a properly operated digester. The average ORP of the digested sludge slightly increased over time with increased air injected from an average of −462 mV at the start to an average of −422 mV and −389 mV during the following first and second month, respectively.

3.3. Case Study #3

Air injected into the digester resulted in a drop in the biogas H_2S concentration also within one week, from about 2300 ppm_v to about 400 ppm_v. During the next six weeks, the air injection rate was kept constant at about 5.1 $m^3\ h^{-1}$, which maintained the H_2S concentration in the biogas at 200–400 ppm_v, with the lowest H_2S concentration measured being 153 ppm_v (Fig. 3). The average H_2S reduction was 87 % ± 3 % when compared with the control digester which was operated similarly but without MA.

The pH of the sludge was 7.25 ± 0.09 without pH control, while the operational temperature was kept on average at 37.5 ± 0.2 °C. The digested sludge during the test period showed a TS content of 2.5 ± 0.03 % (*w*/w) and a VS content that was 72 ± 1.2 % of TS.

On day 16 and day 63 the air injection was stopped for 24 h, which directly resulted in the H_2S concentration in the biogas rapidly reaching the values recorded in the biogas produced in the control digester. Directly after resuming the air inject the following day, the H_2S reduction in the biogas was 67 % after about 6 h and 87 % after 24 h. Another upset test where the sludge gas mixing blower was turned off for 18 h (while air injection was still running) did not negatively affect the H_2S concentration in the biogas (data not shown).

3.4. Sulfur conversion

Dissolved sulfide in the sludge under oxygen-limiting (micro-aerobic) conditions is partially oxidized to elemental sulfur, which seems to be a stable end-product of the MA process [26,21,23]. This was confirmed in all case studies when, after stopping the air injection, the H_2S concentrations returned to their original values (before staring MA), indicating that the produced sulfur product remains stable and no elevated biogas H_2S concentrations can be expected beyond their original values when MA is suddenly stopped.

The sludge feed, the digested product and the biogas were sampled and analyzed for the different sulfur constituents in Case Study #2 to generate a sulfur mass balance (Fig. 4). More than 80 % of the sulfur entered the digester as organically bound sulfur or elemental sulfur, ~10 % as sulfate, ~7.5 % as inorganic sulfide precipitants (with metal ions) and ~1 % as dissolved sulfide. Inside the digester the sulfate is reduced to below detection limits and the total amount of elemental sulfur is increased, while the inorganic sulfide precipitants remain unchanged. The results confirmed that the sulfur that normally left the digester as H_2S via the biogas, does leave the digester via the solid phase of the digested sludge under MA condition when the air is injected into the liquid phase.

Both large-scale tests confirmed the results obtained in bench-scale studies [22,8,23], where MA stimulated the activity of sulfide-oxidizing bacteria, so that elemental sulfur was formed from the partial oxidation of H_2S as per reaction (1).

$$H_2S + 0.5\ O_2 \rightarrow S^0 + H_2O \quad (1)$$

Biological sulfur oxidation is considered as the main mechanism, especially when the air is injected into the sludge liquid broth, but a chemical oxidation potentially catalyzed by the presence of metals (such as iron) may also contribute to the partial oxidation of the dissolved

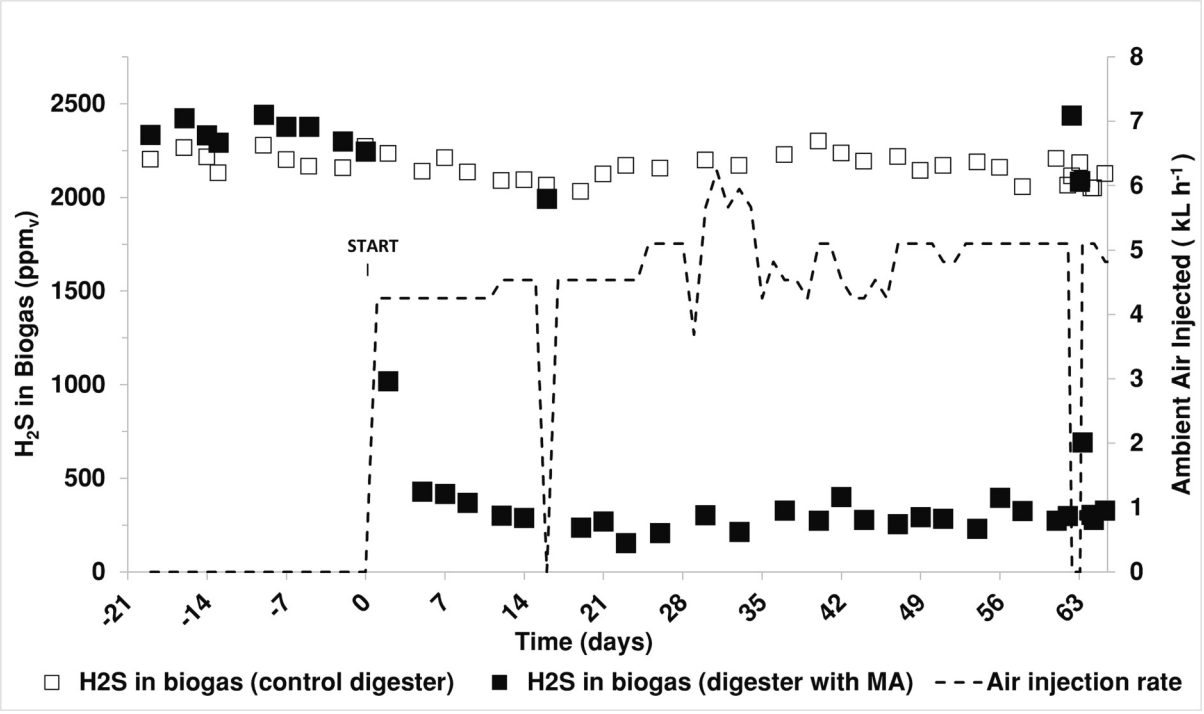


Fig. 3. Hydrogen sulfide concentration in biogas in the digester prior and during micro-aeration in Case Study #3 with air injected in the sludge recirculation pipe. On day 16 and day 63 the MA air injection was stopped for 24 h as part of the evaluation of process reliability.

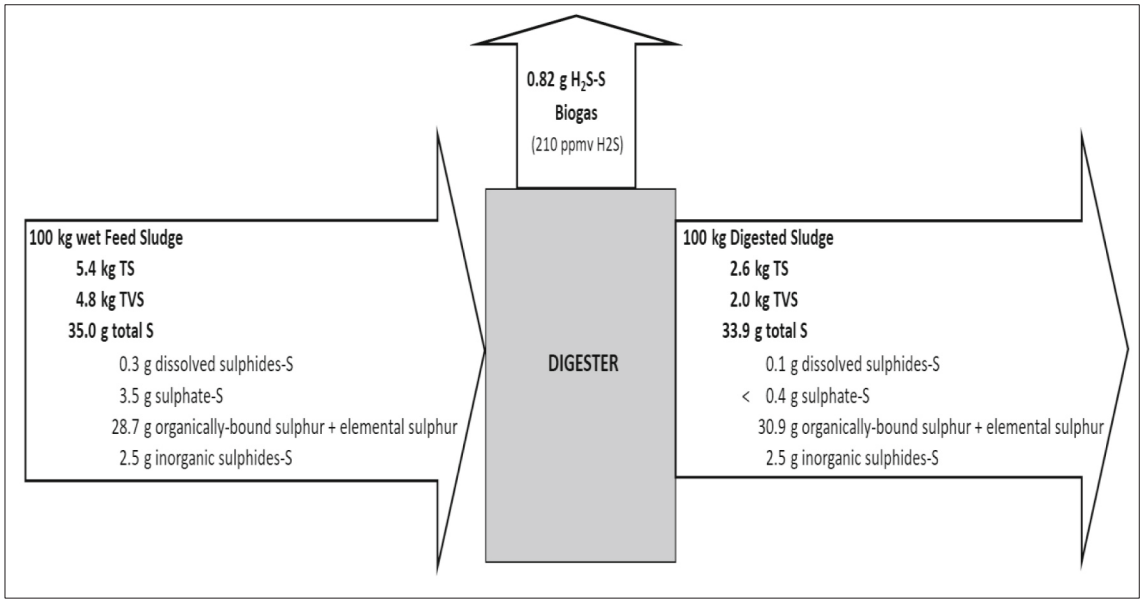


Fig. 4. Sulfur balance of the large-scale micro-aerated digester at day 83 in Case Study #2.

sulfides in the sludge [27,23,28].

4. Discussion

4.1. Critical process parameters

The results of the three large-scale case studies were compared with application data reported in literature to present an overview of known large-scale studies at WRRFs and to critically discuss process parameters. Table 1 shows the applications of MA in large-scale digesters reported in literature and includes the three case studies herein investigated. The overview shows the different large-scale applications including digester size, organic feed material, digester mixing method, typical biogas H_2S concentrations without MA and H_2S reduction obtained under micro-aerobic conditions.

In general, 80 to over 95 % of H_2S reductions have been achieved in large-scale applications, some of which have been in operation for multiple years [11]. The air may be injected in the sludge feed, recirculated sludge or the digester headspace. Most large-scale studies used MA injection in the sludge recirculation pipe used for either sludge heating or sludge mixing. Although sludge mixing may be accomplished through mechanical mixing, for example, sludge recirculation pumps or paddle mixers, it may also involve gas mixing, where biogas is being recirculated from the headspace to the bottom of the digester reactor. Sludge mixing methods during the MA studies mostly rely on sludge recirculation using sludge pumping, although Case Study #2 shows that

Table 1
Example of large-scale digester micro-aeration studies for biogas desulfurization.

Large-scale MA study (#)	Volume digester (m^3)	Digester feedstock[a]	Digester temperature[b]	Digester sludge mixing[c]	Biogas production rate ($Nm^3\ d^{-1}\ m^3\ reactor^{-1}$)	Biogas average H_2S conc. without MA (ppm_v)	Biogas average H_2S reduction with MA (%)	Reference
1	2450	PS + WAS	M	SR	0.61	5100	80	Case Study 1
2	3250	PS + WAS	M	GR	0.50	1125	80[f]	Case Study 2
3	3300	PS + WAS	M	SR	0.83	2173	87	Case Study 3
4	2100	PS + WAS	M	SR	0.40	5652	99	Jenicek et al.
5	30 000	PS + WAS	M	SR	0.32	1963	74	[19]
6	3000	PS + WAS	M	SR	0.30	3268	99	
7	3200	PS + WAS	M	SR	0.31	2851	95	
8	1900	PS + WAS	M	SR	0.27	664	95	
9	2600	PS + WAS	M	SR	0.37	575	96	
10	1600	PS + WAS	M	SR	0.47	2767	88	
11	2400	PS + WAS	M	SR	0.36	1840	90	Diaz et al. [29]
12	4000	PS + WAS + SSO	T	SR	0.14[d]	1850	99	Giordano et al. [12]
13	4500	PS	M	PM	0.94/0.39[e]	578/179[e]	40–80[g]	Cesca et al. [30]

[a] PS = primary sludge; WAS = waste activated sludge; SSO = source segregated organics (i.e., domestic food waste).
[b] M = mesophilic; T = thermophilic.
[c] SR = sludge recirculation; GR = gas recirculation; PM = paddle mixers.
[d] Average production of 3 digesters in series.
[e] Average of two primary digesters and one secondary digester, respectively.
[f] 80 % reduction when the MA air was injected in the sludge recirculation stream.
[g] The H_2S reduction is highly variable and was sometimes over 90 %. However average H_2S reduction was about 40 % in primary digesters and about 80 % in the secondary digester.

MA in digesters with gas mixing was also proven to be effective. Although most of the tested digesters were operated under mesophilic conditions, MA was also proven recently to be effective in digesters operated under thermophilic conditions [19].

Table 2 shows details of the digester sludge loading rates and MA injection methods including air injection rates for the different large-scale MA studies mentioned in Table 1. Most large-scale MA applications involve air injection into the sludge liquid, but injection in the digester headspace has also been proven to be effective in large-scale applications [29,12]. Air injection in the bottom of the digester tested in Case Study #2 was less effective than when the air was injected in the sludge recirculation stream similarly to the study described by Cesca and co-workers [30]. This was postulated to be caused by the lower oxygen transfer efficiency rather than by the MA injection location.

Different bench-scale studies have been performed to determine the optimal oxygen dosage rate at low and high H_2S concentrations, mainly using oxygen dosage in the headspace of the digester reactor. The optimal sulfide removal efficiency can be expected to be about 1 % (*v*/v) of pure O_2 per volume of biogas when treating medium to high H_2S concentrations (i.e., 3300 ppm_v) [18]. Nevertheless, the amount of air to be dosed seems to be dependent on multiple factors and not only the biogas H_2S concentration. From the large-scale MA studies reported herein, the following general guidelines for large-scale micro-aerobic digestion were obtained:

- Air dosing rate (ambient air) required is typically between 1 and 8 % of the biogas flow rate.
- Air dosing rate (ambient air) required is typically between 0.1 and 1.75 L h^{-1} per m^3 reactor.
- Air dosing should provide at least 2.5 times the stoichiometric amount of O_2 required for the partial H_2S oxidation to elemental sulfur.
- More than 90 % H_2S removal can be obtained typically when the specific biogas production rate of the digester is up to 0.4 Nm^3 biogas per day per m^3 reactor. Higher digestion activities seem to result in a lower biogas desulfurization efficiency as illustrated in Fig. 5. The digestion activity is affected by organic loading rate and hydraulic retention time which could be also parameters influencing desulfurization efficiency. Similarly, digesters fed with only primary sludge are more active with a higher competition for oxygen from the injected air, which may lower the biogas desulfurization efficiency (see Table 1, Large-scale MA Study #13). The oxygen transfer rate efficiency of the MA air injection could also be affected by the volumetric biogas production rate (i.e., gas bubble formation rate in sludge liquid).
- Keeping the oxygen dosage below 17 kg O_2 per ton VS should not upset the digestion process.
- While MA can be applied in digesters using liquid pump mixing, gas mixing and paddle mixing, the sludge mixing intensity and method may be highly critical in relation to MA performance and/or the MA injection point. Discontinuous mixing (1 h on/1 h off as tested in Case Study #1) showed to significantly impact the MA performance (see Section 3.2 herein). Moreover, recent MA testing in a full-scale donut-type of digester showed that biogas desulfurization was not good possible when injecting MA air at the heat sludge recirculation pipe which recirculated the sludge in the outer part of the digester, while fresh sludge was fed to the inner part of the digester (data not shown here).
- Air injection seems best relatively close to the feedstock injection point.

4.2. Microbial contemplations

MA provides trace amounts of oxygen for facultative microbes that consume and partially reduce oxygen levels [14] in synergistic interactions between facultative and anaerobic microorganisms [26]. The concentration of dissolved oxygen present in the sludge is negligible as it is quickly and completely consumed during the MA process. Only part of the injected oxygen is utilized for the oxidation of dissolved sulfur while the rest is utilized by other facultative microbes to oxidize readily biodegradable organic matter or leaves the digester as residual oxygen in the biogas. It has been shown that the overall digestion process can be improved by MA due to a more diverse microbial community that may evolve during MA, which may also enhance the hydrolysis processes (typically the rate limiting step in anaerobic digesters). In a micro-aerated anaerobic process, an environment may be created that maintains opportunities for both anaerobic and micro-aerobic facultative microbes. Some strict anaerobes can survive and thrive without any inhibition when the injected oxygen levels are low [31]. In addition, under certain microaerobic conditions, several strict anaerobes have

Table 2
Details of the process conditions of the large-scale digester micro-aeration studies.

Large-scale MA study (#)	VS loading	HRT[a]	Micro-aeration air injection method[b]	Micro-aeration air dose rate			
	kg_{VS} m^{-3} d^{-1}	d		L h^{-1} per m^3 digester	% (v/v) air of biogas produced	Ratio dosed O_2 to O_2 required[d] for H_2S partial oxidation	kg O_2 ton VS^{-1}
1	2.1	22	Air in SR stream (AN)	5.4[e]	21[e]	18	16.4
2	3.3	16	Air in SR stream (AN)	1.75	8	28	3.8
			Air in sludge digester (OP)	1.75			
3	1.4	30	Air in SR stream (AN)	1.5	4.5	8.7	7.8
4	–	–	Air in SR stream (OP)	0.6	3.5	2.6	–
5			Air in SR stream (OP)	0.2	1.5	3.2	
6			Air in SR stream (OP)	0.4	3.2	4.1	
7			Air in SR stream (OP)	0.3	2.4	3.5	
8			Air in SR stream (OP)	0.1	1.3	7.9	
9			Air in SR stream (OP)	0.1	0.9	6.3	
10			Air in SR stream (OP)	0.6	3.2	4.9	
11	0.5	45	Oxygen (93 % v/v) in HS digester	0.1	0.8	8.2	7.6
12	1.9	58[c]	Air in HS digester	0.7	3.8	8.6	2.1
13	2.3	16.5	Air in sludge prim. digester (OP)	6.0	5.4	37[f]	16.9

[a] HTR = hydraulic retention time of the organic feed material in the digester.
[b] OP = open pipe; HS = head space; AN = air injection nozzles.
[c] The total HRT of three digesters in series.
[d] Stoichiometric oxygen requirement of 0.5 mol O_2 per 1 mol H_2S as per formula (1).
[e] Dose rate may not be representative due to partial accumulation of debris on the test nozzles.
[f] Relatively high due to type of MA injection (duck beak nozzles on bottom reactor) and type of feed stock (primary sludge only).

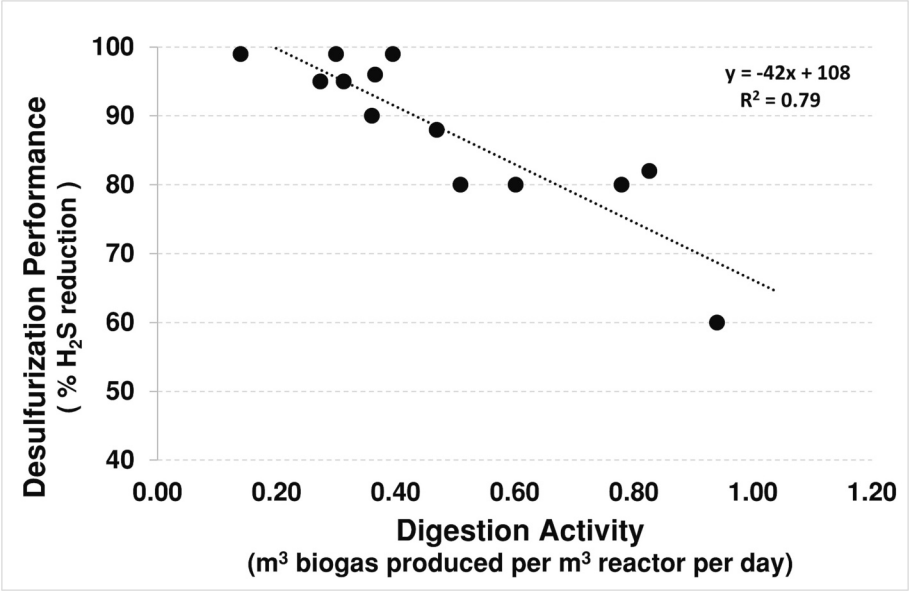


Fig. 5. Desulfurization performance versus specific biogas production rate in large-scale micro-aerobic digesters.

shown that they can produce enzymes to adapt to oxidative environments [32].

During anaerobic digestion several interdependent biochemical processes are required to produce biogas, which are sensitive, especially to high organic loading rates. The anaerobic process involves a series of interdependent biochemical steps i.e., hydrolysis, acidogenesis, syntrophic acetogenesis, and methanogenesis carried out by diverse microbial groups. These interactions between microbes enable organic material degradation including some thermodynamically less favorable reactions [14]. Anaerobic digestion can be an unstable process, due to the disparity between the acetogenic bacteria and methanogens, resulting in VFA accumulation, or other process instability often resulting in foam formation. Testing of typical potential up-set scenarios of the MA process in the digesters in our three full-scale studies did not negatively affect the anaerobic digestion process. The VFA/alkalinity ratio as well as the biogas production stayed stable in these upset tests. Micro-aeration may not only promote sulfide oxidation to reduce H_2S in biogas but may also enhance the hydrolysis and fermentation of organic matter. Micro-aerobic conditions may facilitate aerobic VFA oxidation by heterotrophs and may be a strategy to mitigate the inhibitory effects caused by accumulating intermediates, thus facilitating the overall stability of anaerobic digestion processes. The full-scale tests herein confirm bench-scale experiments that micro-aerobic conditions may therefore also be a simple solution to enhance anaerobic process stability [13,15,21,33].

Controlling or optimizing the trace amount of oxygen in the digester could be paramount. The ORP can sense very small changes in oxygen concentrations and could be a control parameter for MA. Monitoring the ORP as a parameter to regulate micro-oxygen injection into anaerobic digesters for reducing H_2S concentration in biogas has been proposed by Nghiem and co-workers [24]. ORP varies linearly with the logarithmic of the oxygen concentrations. Methane production will occur when the ORP is −200 mV or less and acid formation (fermentation with potential volatile fatty acid accumulation inhibiting methanogenic activity) will mainly occur when the ORP ranges between −75 and −225 mV, while micro-aerobic digestion can be defined by the ORP range between 0 mV and −300 mV [14].

In Case Study #2, ORP-measurements were performed after optimization of the type of probe and location of the probe. Successful relatively stable measurements were obtained for several weeks with an online ORP meter (HACH) tested in the bypass line of the sludge recirculation pump. The results of the online measurements are shown in Fig. 6. Fig. 6a shows the results of the ORP measurement taken every 15 min, Fig. 6b shows the ORP measurement results averaged over 1 h, while Fig. 6c shows the ORP measurements averaged over a 4-hour period. The 1-hour and 4-hour average would average out the measurements taken every 15 min to reduce the 'noise' within the results of this very sensitive measurement. A 4-hour average data point was found to be sufficient to potentially provide a feedback loop control to the amount of air injected as the digestion process is relatively slow and with biogas being recirculated within the digested every ~4 h on average. Despite relatively stable ORP measurements were obtained with the Hach ORP transmitter installed in the sludge pump bypass line, the ORP sensor did not provided a direct response when air injection rates were changed. In addition, the ORP probe still had to be cleaned frequently as debris slowly accumulated on the probe, which render it not reliable enough to serve as a control parameter for the MA process. Rather being a control parameter the ORP may still be a useful monitoring parameter for the MA process, without any control function. All three Case Studies showed that, when the digester is fed and operated in a constant manner, a constant air injection rate during long-term operation can provide a stable H_2S desulfurization process without any direct control.

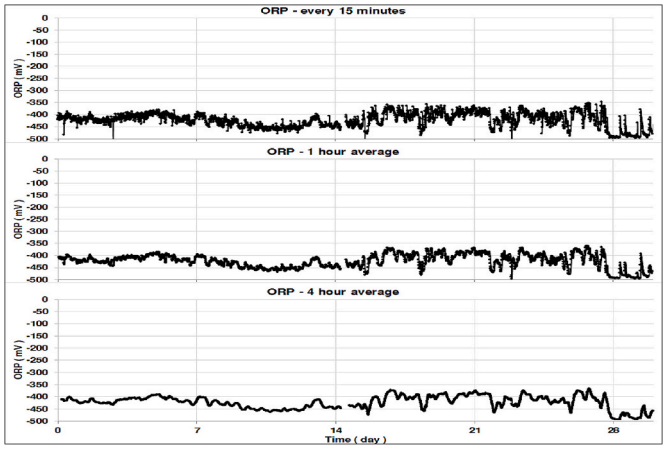


Fig. 6. ORP measurement at large-scale micro-aeration testing during Case Study #2.

Several typical digestion or micro-aeration process changes or upset scenarios were investigated to gain an improved understanding of the reliability of the micro-aerobic digestion process. A sudden stop of the air injection (all Case Studies) showed that the H_2S levels in biogas rapidly returned to their original values (before MA). When re-starting air injection the H_2S concentration dropped within a day to its value before stopping the air injection. When air injection rate was overdosed (Case Study #1), the VFA/Alkalinity ratio increased first, followed by a slight drop in pH the next day and a reduced methane production during the following days. While the air injection was left off, within 3 days the pH recovered to its original value, while full recovery of the methane production took longer: about 5 days to 90 % and about 10 days to 100 % recovery. A 2-hour sludge recycle pump failure followed by a digester feed pump stop for 18 h (Case Study #3) did not cause a decrease in the efficiency of biogas desulfurization. The understanding of the reliability of the process can provide deeper understanding of the MA process and may provide valuable input into the design of permanent large-scale MA in terms of instrumentation, monitoring, and automated process controls.

4.3. Safety considerations

The AD process is typically strictly anaerobic because of the potential inhibition in growth and activity of methanogens. Moreover, anaerobic conditions are maintained as the addition of oxygen to biogas has a risk of explosion at elevated oxygen concentrations. The upper explosion limit (UEL, fuel-rich) and lower explosion limits (LEL, fuel-lean) are referred to as the mole fractions of the biogas in air at which flame propagation after spark ignition no longer occurs. The range between the lower and upper explosion limits is known as explosion range, which depends on the biogas composition, and is typically between 5 % (*v*/v) and 25 % (v/v). Methane leaking from process units into the ambient air environment should be the main concern regarding the formation of an explosive atmosphere at WRRFs. Micro-aerobic digestion on the other hand operates at an oxygen-to-biogas ratio robustly lower than that required to reach the upper flammability limit of biogas/air mixtures. Nevertheless, control/safety strategies need to be implemented to maintain safe operations of large-scale digesters operated under micro-aerobic conditions.

To eliminate any risks both intrinsic measures and extrinsic measures are proposed here. Intrinsic measures are features inherent of the process which can't be changed during its operation and may be an orifice and air fuse as part of the MA air supply system. Extrinsic measures are additional features to the process controls and may be air injection control interlocks with feedback from equipment/instruments. The control of air injection stops as it interlocks for example with the sludge recirculation flow interruption, any gas recirculation interruption, any sludge feed interruptions and/or too high oxygen concentrations in biogas. While the extrinsic measures should be robust controls, the intrinsic are the basic line of defense against unsafe conditions independent on instruments and instrument feedback loops, and independent of power availability.

4.4. High-level comparison with conventional biogas desulfurization technologies

Table 3 shows a high-level comparison of MA with conventional biogas desulfurization methods including ferric chloride dosing and end-of-pipe systems such as dry media adsorbents, iron coated media (e.g., iron sponge) or wet scrubbing systems. The comparison shows some of the important considerations when using the different methods, advantages and disadvantages and general capital and operational costs based on existing facilities.

The potential capital and operational cost savings that MA technology could provide are substantial. The capital cost for an end-of-pipe biogas desulfurization technology varies depending on the biogas flow rate and inlet H_2S concentration. This can be well over one million dollars installed for a typical mid-size WRRF - while the capital cost of a MA system is estimated to be about $250,000. These are in line with the economic analysis of microaerobic in large-scale sludge digesters as described by Diaz and c-workers [9]. Moreover, the operational cost of MA is negligible compared to end-of-pipe technologies, which may require input of chemicals such as caustic, activated carbon or iron-based material, etc. [34], while MA eliminates the disposal of the typical end-products for these end-of-pipe technologies. This includes sulfuric acid, elemental sulfur slurry or spent iron oxide–based adsorbents and extra sludge requiring disposal in the case of ferric chloride dosing.

Conventional measures are typically in place in digestion facilities at wastewater facilities to reduce the H_2S in biogas, but they require significant ongoing costs and could be improved in terms of sustainability (e.g., iron dosing requiring ongoing input of chemicals or end-of-pipe systems like coated dry media system such as iron sponge or impregnated activated carbon media requiring frequent replacement and disposal of the spent media typical to landfill). In many of these cases can MA provide large benefits in terms of cost savings and environmental benefits (less chemical usage, less input of raw materials, lower energy cost, lower carbon footprint). In those cases that MA provides cannot reduce the H_2S concentrations low enough, the existing systems can still be used as polishing while MA is providing the bulk of the effort in terms of biogas desulfurization.

MA can provide high biogas desulfurization efficiencies (>95 %) as illustrated in Fig. 5, but microbial activity inside the digester (expressed in the volumetric biogas formation rate) is relevant, which can provide guidance for what efficiencies are feasible with MA (or could also be used as guide to design or set digester loading rates for when MA is to be implemented). Further polishing of the biogas may, in certain cases, still be required when applying MA depending on the H_2S concentration level in the biogas and the downstream processing of the biogas (e.g., upgrading to biomethane).

Table 3
General consideration of conventional biogas desulfurization technologies compared with Micro-aeration.

	Ferric or ferrous dosing to feed sludge	Iron oxide coated dry media	High-capacity activated carbon	Wet scrubbers (water/chemical/biological)	Micro-aeration
Capital cost	Low	Low	Low	High	Low
Operating costs ($ per kg H_2S removed)	High (16–25)	High (10–14)	High (8–18)	Low–medium (1–8)	Low (<0.5)
Operator attention	Moderate	Low	Low	High	Low
Operational considerations	Safety (pH ~ 2)/some increase in sludge volume	Frequent media changed out required	Frequent media changed out required	Control challenges/could involve hazardous chemicals	Safety/slight increase in sludge volume
Water input/leachate discharge	None	Inlet gas must be fully saturated	Moisture must be removed from inlet gas	Requires soft water/creates sulfur slurry	None
Disposal considerations	None	Non-hazardous landfill (when not acidic)	Hazardous landfill (typically acidic)	Sulfur slurry	None
Performance consistency	Poor consistency	Good	Good, but quickly results in breakthrough	Poor consistency	May need polishing

A potential drawback of micro-aeration could be the risk of diluting the methane concentration excessively such that the biogas is too lean for proper combustion. Alternatively, concentrated or pure oxygen rather than air may be used as demonstrated by Diaz and coworkers [29], especially when biogas is upgraded to biomethane, because nitrogen is an inert gas very difficult to remove from the biogas during upgrading to biomethane.

The accumulation of elemental sulfur in the digester headspace or downstream process units such as flame arrestors might potentially cause operational problems. Long-term (one year or more) operation of MA can result in elemental sulfur deposition on flame arrestors requiring more frequent cleaning as have been observed in the digesters described in Case Study #2 after the MA process for about 2 years after the study. Cleaning and maintenance would demand a dual arrangement at any pressure relief valve (PRV) and flame arrestors. This drawback of MA can be managed by installing dual PRV arrangements when micro-aeration is implemented and should be combined with additional instrumentation (especially air injection monitoring and oxygen monitoring in the biogas).

5. Conclusions

Micro-aerobic digestion studies in three different large-scale digesters at WRRFs were conducted and the results support bench-scale research performed to date that micro-aeration technology holds potential as an effective and more sustainable biogas desulfurization method compared to conventional biogas desulfurization technologies. A sulfur balance over a full-scale sludge digestion process was successfully obtained. This, together with micro-aerobic process interruption tests showed that a stable end-product (elemental sulfur) is formed in the liquid phase, which leaves the digester via the solid phase of the digested sludge when air is injected into the liquid phase. The air injection rate and the air injection location as well as the volumetric biogas production rate showed to be important parameters affecting the micro-aerobic desulfurization performance in large-scale digesters. The need to a process parameter for automated control (e.g., ORP) seems not requisite for the MA process in large-scale sludge digesters at WRRFs. Reliability testing of typical upset scenarios and long-term micro-aerobic digester operation provided deeper insights in the MA process that could also provide valuable input to what operational, maintenance and safety measures should be considered when applying micro-aeration in large-scale digesters.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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